Engineering Mechanics 1 · Unit_02 · UNIT 2

DRAFT — UE26ME142A · Unit_02 · UNIT 2 · UE26ME142A Unit 2 Class 15 AVSummary

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

UE26ME142A Unit 2 Class 15 AVSummary

1 UE26ME142A Unit 2 Class 15 AVSummary

Study focus: UE26ME142A_Unit 2_Class 15_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000022 · EV0000023

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Svend Kitish from Civil Engineering under Department of Science and Humanities, PS University.
- This is class number nineteen.
- And this is class number two.
- Under unit two.
- Under unit two, in class one, that is class number eighteen, we had discussed about the definition of equilibrium.
- What are the equity that need to be considered for a body to be in equilibrium?
- Then we have looked at few of the concepts or definitions, or what is that is required for us to know to understand equilibrium.
- How do we draw a free body diagram?
- And various different.
- What are the different types of supports that we should know?
- And what are the reactions that is going to be acting because of those supports were written in the first class of unit 2.
- Today we will be continuing in this class further.
- The first one is collinear.
- In this case, it is taken that the forces is acting along X axis.
- If this equation is satisfied, then we can easily say.
- That the body is in equilibrium.
- We don't have to look at anything else.

- Because there is no force acting perpendicular to it.
- Then movement about any point is equal to zero, because if the net force acting is zero.
- Then that would not produce any movement except if it is a couple.
- But in this case, since all the forces are acting along the same axis, that would not produce any couple.
- I hope that is clear.
- To specify whether the body is in equilibrium or not.
- If the algebra sum of all the forces is equal to zero, the body will be in equilibrium.
- The second category of the force system is concurrent at a point.
- What do you mean by concurrent?
- The point of application of all the forces when extended would meet at the same point.
- Like this, whatever is indicated.
- And if I resolve these forces along this axis X and Y, and if I could get the algebra sum of all the forces along X to be equal to 0, and the algebraic sum of all the forces along y to be equal to 0.
- That means the result is 0.
- If this is proved, then I don't have to go and take the other equation that is sigma m equals to 0.
- The body is in equilibrium.
- That is because the point of application of all these forces meet at the same point O.
- But the resultant is equal to zero because summation of all the forces along X is 0, along Y is also 0.
- That is only when there is concurrent, the forces are concurrent.
- The next category is a parallel force system.
- There are four forces acting.

Example or application from the lecture

- For example, a free body diagram.
- Tables were given where how to draw the free body diagram was explained and there were some diagrams which was explained which was wrong and how do we correct that also was explained.
- For example, summation of all the forces perpendicular to x-axis.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

Core explanation · 05:00–09:59

Yes, the displacement may not be there, but it can produce a moment.

Key details

- The forces or the summation of the moment of all the forces about the z-axis about any point should be equal to zero.
- If that is also satisfied, then this body would be in equilibrium.
- The next category is a general category where you have several force system.

- You can also have an applied moment.
- The forces when it is extended may not meet at the same point.
- In such case, we have to have all these three equations of equilibrium.
- Which should be satisfied, that is algebraic sum of all the forces acting along yes.
- Algebic sum of all the forces acting along Y?
- And the moment about the z-axis about any point should be equal to zero.
- If it satisfies all these three equations, then we can say that this body is in equilibrium.
- the component of all the forces about the algebraic sum of the components of all the forces about x axis should be zero.
- Then it means that there is no displacement along X.
- The algebraic sum of all the components of all the forces about y axis is 0, then I can also say that the displacement along y is 0.
- And if the movement of all the forces and the couple or the moment.
- About any point along the z axis is equal to zero, then there is no rotation.
- And these three equations are called as the equations of equilibrium for a body which is considered to be a two dimensional.
- We are not looking at three dimensional.
- If it is three dimensional, we will have additional equations that needs to be proved or additional things that needs to be proved.
- In addition to all these three, we will have another three equations.
- That is algebraic sum of the components of all the forces about the z axis also should be equal to zero.
- Then algebraic sum of the moments about x axis should be zero.
- Then algebraic sum of all the moments about the y axis also should be equal to zero.
- And six will be the equation of equilibrium for a three dimensional.
- But if it is two dimensional and if we take the body to be in X and Y axis, then these are the three equations of equilibrium, which has to be satisfied for us to say that the body is in equilibrium.
- Next continuation of this.
- If we have a two force members, how do we say whether the body is in equilibrium or not?
- If it is two force members, if it is collinear, and if one force is opposite to the other, plus.
- If the magnitude of the force is the same, then we can easily say that the body is in equilibrium.
- That is, if I take the algebraic sum of all the forces along this axis, okay, then I can easily say that the body is in equilibrium.
- Next, if I take a three force system, something like this, which is concurrent, then if this follows the law of triangle of forces as indicated here, then I can easily say that the body is in equilibrium.
- If you take the two forces on the adjacent sides of a parallelogram, if the other force forms the other side of the parallelogram, other side of the triangle taken in the opposite order, then we can easily say that the body is in equilibrium.
- Along this axis, and I also find that the movement about point A is zero.

Example or application from the lecture

- Say for example I take a body, there are several forces acting, and if I find that there is a result, if it exists, if there is a result and that is acting.

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Core explanation · 10:00–14:59

If we take an axis perpendicular to the resultant force if it exists, okay, and if the summation of all the forces along this axis is 0.

Key details

- And we also find that the moment about the point A, which is on this result and axis to be equal to zero.
- The movement about any point B where A and B is not perpendicular to R, if that is also equal to 0, then I can easily say that the body is in equilibrium.
- Other equations of equilibrium for such thing is your summation of all the forces about the x-axis should be equal to zero.
- The moment if a resultant exists.
- Along that line of action of the resultant, M A should be equal to 0.
- And also, if you take any point such that AB is not perpendicular to R, and if the movement about B is also equal to 0.
- Then I can easily say that the body is in equilibrium.
- Another condition that we take that if there exists a result, something like this, and if we found that the movement about point A is 0.
- And if I take some other point B and even there the moment because of all the forces is zero, that means that the resultant has to act along this A B line only.
- In addition to this, we take another point C where this AB or B C are not perpendicular to R, either this or this, and if we find that the moment of this about C is also 0.
- Then also we can say that the body is in equilibrium.
- So first thing is I have told you that sigma f x should be equal to 0, sigma f y should be equal to 0, and the sigma about the z axis about any point should be equal to 0.
- These are equations of equilibrium.
- Other than this, if I don't want to take, then I can also take this condition where I will be proving sigma fx equal to 0, sigma is equal to 0, where a is taken as the point along the line of action of the resultant if it exists.
- And also, if I can prove that sigma m b is equal to 0, where AB is not perpendicular to the resultant, if it exists, then I can say that the body is in equilibrium.
- Or if I take something like this, if there exists a resultant, and if I find that.
- About the point the moment is 0 where A is on the line of action of the resultant, and also if I take some other point B where AB is not perpendicular to R, and if I find that sigma M B is also 0, then your resultant if it takes it, it has to act along AB.
- And also I will take a point C where A C or A B C are not perpendicular to this resultant or this resultant.
- And if I find that movement about C because of this is also zero.

- Then I can easily say that the body is in equilibrium.
- This is these are the alternate equations of equipment that I can take.
- If I could resolve all the forces along X and Y, which I could do, then this constraint or this support will balance all the forces acting along the X axis, then there won't be any displacement along X.
- Then if the all the forces, the component of the force that is acting along y, if this support or constraint balances along with this of all this vertical force that is acting on the body, then there won't be any displacement along the y-axis.
- Now, if I take this point A, and if the moment about this point, because of all the forces that is acting along with this support reaction value.

Example or application from the lecture

- Consider here or the equations of equilibrium.
- Next, let's look at what are the fixity and adequate constraints that is required for me to say that the body is in equilibrium.
- Now there is an example that is taken here.

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EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

Core explanation · 15:00–19:59

Will be equal to zero.

Key details

- That means there is no rotation that is going to occur.
- Along with the support reaction values.
- Then I can say that it is it has got a complete fixity and the constraint is also sufficient for complete fixity.
- Or for the equations of equilibrium to be valid.
- This may prevent the translatory motion along X, this may prevent the rotatory motion translatory motion along Y, but since this is acting along the line of action passes through A itself.
- Now moment about A because of this would not come into picture.
- So that means this body now can or able to rotate.
- Along this three, all this three is producing the displacement along X to be 0.
- Whereas when I apply a force here, vertical force, there is no constraint.
- It does not prevent it from displacing along Y.
- Or a partial constraint because it allows the displacement to occur a long way.
- Along with this, there is one more support here.
- Whether it passes through A or it does not pass through A, it does not matter.
- But if you compare this diagram and this diagram, only this support, this support and this support should have been enough for complete fixity and adequate constraint.
- But here we have excess fixity.
- One extra support is there.

- Whatever is required for equilibrium.
- More than that, we have provided a support.
- There is one redundant constraint that is existing.
- before we get into the problem solving mode.
- If it is excessive then you will have a redundant constraint.
- Then the sufficient should be sufficient so that there is no translatory motion or rotatory motion.
- If anyone is possible to occur, then we cannot say that the body is in equilibrium.
- And how do I how can how I can prove that?
- By using the or if it doesn't satisfy the equations of equilibrium, I can easily say that there is no adequate.
- constraint or I can say that it is partially constrained.
- We will continue in the next class that is class number 20.
- That would be the class number three under unit two.
- I am Doctor Sv Vengtesh.
- We will meet in the next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000034 · EV0000035 · EV0000036 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
UE26ME142A_Unit 2_Class 15_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/015_UE26ME142A_Unit_2_Class_15_AVSummary__29e4398836d93577.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 15_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/015_UE26ME142A_Unit_2_Class_15_AVSummary__29e4398836d93577.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 15_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/015_UE26ME142A_Unit_2_Class_15_AVSummary__29e4398836d93577.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 15_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/015_UE26ME142A_Unit_2_Class_15_AVSummary__29e4398836d93577.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.slides.md have no readable `.slides.md` text; their original page images are included.
2. 4 rendered pages in 015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.slides.md have no readable `.slides.md` text; their original page images are included.
4. 4 rendered pages in 017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.slides.md have no readable `.slides.md` text; their original page images are included.
5. 3 rendered pages in 018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.slides.md have no readable `.slides.md` text; their original page images are included.
6. 3 rendered pages in 019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.slides.md have no readable `.slides.md` text; their original page images are included.
7. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.slides.md have no readable `.slides.md` text; their original page images are included.

9. 3 rendered pages in 019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.slides.md have no readable `.slides.md` text; their original page images are included.
10. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.slides.md have no readable `.slides.md` text; their original page images are included.
11. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.slides.md have no readable `.slides.md` text; their original page images are included.
12. 3 rendered pages in 021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md have no readable `.slides.md` text; their original page images are included.
13. 2 rendered pages in 022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.slides.md have no readable `.slides.md` text; their original page images are included.
14. 3 rendered pages in 023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.slides.md have no readable `.slides.md` text; their original page images are included.
16. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
17. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
18. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
19. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
20. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00006	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare...
SRC00007	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00008	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare...
SRC00009	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare...

C Revision and retrieval practice